

Claim. Let S be a finite-dimensional local commutative k -algebra, I a finite S -module, and $i : S \hookrightarrow I$ injective. Let $A := \text{Sym}_S(I) \otimes_{S[x]} S \cong \text{Sym}_S(I)/(i(1) - 1)$ (where $x \mapsto i(1)$), and assume the induced map $\phi : I \rightarrow A$ is injective. Then there is an ideal $K \subset \text{Sym}_S(I)$ such that $\text{Sym}_S(I)/K$ is finite-dimensional over k and $I \rightarrow \text{Sym}_S(I)/K$ is injective.

Proof (concise). Set $V := \phi(I) \subset A$; this is finite-dimensional over k . Since I is finite over S and S is finite over k , A is a finitely generated k -algebra, hence Noetherian.

(1) *Separate V by finitely many localizations.* For $0 \neq v \in V$, pick a maximal ideal $\mathfrak{m}$ containing $\text{Ann}_A(v)$. Then $v/1 \neq 0$ in $A_{\mathfrak{m}}$ (else $sv = 0$ for some $s \notin \mathfrak{m}$, forcing $s \in \text{Ann}(v) \subset \mathfrak{m}$). Iterating on $\ker(V \rightarrow A_{\mathfrak{m}})$ yields maximal ideals $\mathfrak{m}_1, \dots, \mathfrak{m}_r$ such that $V \rightarrow \prod_{j=1}^r A_{\mathfrak{m}_j}$ is injective.

(2) *Replace by Artinian quotients.* In each Noetherian local ring $A_{\mathfrak{m}_j}$, Krull intersection gives $\bigcap_{n \geq 1} \mathfrak{m}_j^n A_{\mathfrak{m}_j} = 0$, so $\ker(V \rightarrow A_{\mathfrak{m}_j}) = \bigcap_{n \geq 1} \ker(V \rightarrow A_{\mathfrak{m}_j}/\mathfrak{m}_j^n A_{\mathfrak{m}_j})$. The kernels stabilize (finite-dimensional V), so choose N_j with $\ker(V \rightarrow A_{\mathfrak{m}_j}/\mathfrak{m}_j^{N_j} A_{\mathfrak{m}_j}) = \ker(V \rightarrow A_{\mathfrak{m}_j})$. Hence $V \rightarrow \prod_j A_{\mathfrak{m}_j}/\mathfrak{m}_j^{N_j} A_{\mathfrak{m}_j}$ is injective.

(3) *Cofinite ideal avoiding V .* Let $J := \bigcap_{j=1}^r \mathfrak{m}_j^{N_j} \subset A$. Then $V \cap J = 0$, so $I \rightarrow A/J$ is injective. Also $A/J \hookrightarrow \prod_j A/\mathfrak{m}_j^{N_j}$, and each $A/\mathfrak{m}_j^{N_j}$ is finite-dimensional over k (f.g. k -algebra quotient by a power of a maximal ideal), hence so is A/J .

Finally, let $\pi : \text{Sym}_S(I) \twoheadrightarrow A$ and set $K := \pi^{-1}(J)$. Then $\text{Sym}_S(I)/K \cong A/J$ is finite-dimensional over k , and $I \rightarrow \text{Sym}_S(I)/K$ is injective (identified with $I \rightarrow A/J$). $\square$